Symmetric Diffeomorphic Image Registration on Riemannian Manifolds: Eulerian Normalization, Time-Varying Velocity Fields, and Sobolev-Preconditioned Optimization on the 90-Pair Mindboggle Cohort

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Abstract

Spatial correspondence estimation between structural medical images requires coordinate transformations that preserve topological integrity, prevent non-invertible singularities, and capture high-frequency anatomical boundaries. While the Symmetric Normalization (`SyN`) framework represents the classical reference for diffeomorphic registration, conventional implementations rely on CPU-bound spatial stepping loops and isotropic Gaussian filtering that attenuate fine cortical boundary gradients. In this work, we present a unified mathematical and algorithmic formulation of symmetric diffeomorphic registration and Large Deformation Diffeomorphic Metric Mapping (LDDMM) with continuous Time-Varying Velocity Fields (TVF), accelerated via tensor automatic-differentiation paradigms.

We address fundamental theoretical challenges in variational diffeomorphic optimization: 1. **Asymptotic Variance Singularities in Local Normalized Cross-Correlation (LNCC)**: We prove that unregularized analytical L^2 gradients of LNCC diverge as $\mathcal{O}(\mathrm{Var}^{-1/2})$ in homogeneous intensity regions, inducing non-physical coordinate forces, and establish a safe variance floor $\mathrm{Var}_{\mathrm{safe}} = \max(\mathrm{Var}(I), \epsilon)$ that guarantees bounded gradient dynamics. 2. **Smooth Lie Group SO(3) Mapping**: We establish first-order Taylor continuity at the identity of the Lie algebra $\mathfrak{so}(3)$ exponential map, eliminating zero-gradient lockup during rigid initialization. 3. **Riemannian Sobolev-Preconditioned Optimization (`SobolevAdam`)**: We demonstrate that pointwise adaptive moment optimizers (such as Adam) destroy the spatial regularity of velocity fields in infinite-dimensional function spaces by scaling noise up to unit magnitude in flat domains. We introduce **`SobolevAdam`**, which preconditions parameter updates with the Sobolev-Riemannian Green's operator $\mathcal{G}_{\mathrm{Sobolev}} = (I - \alpha\Delta)^{-s}$, strictly guaranteeing positive Jacobian determinants ($\det(J) > 0$) across continuous velocity flows. 4. **Antisymmetric Velocity Projection & Fréchet Midpoint Anchoring**: We formulate the unique orthogonal projection of forward and backward velocity update pairs onto the antisymmetric subspace, eliminating common-mode translation drift and anchoring the symmetric geodesic midpoint strictly at the Fréchet mean. 5. **Deterministic Multi-Start Manifold Search**: We introduce

a deterministic 18-cone Lie algebra perturbation grid with foreground union-masked Mutual Information scoring, eliminating angular basin entrapment during affine initialization.

We validate the framework on the standardized **90-pair Mindboggle-101 benchmark** (40 intra-subject longitudinal pairs and 50 inter-subject cross-site pairs) evaluated against ground-truth DKT31 cortical label maps sharing locked canonical affine baselines (`0.3499` baseline DICE): - **Eulerian SyN (Gaussian)** achieves `0.6382` **Mean Symmetric Cortical DICE** (+1.66% over classical ANTs SyN `0.6216`, with an 88/90 win rate). - **Eulerian SyN (Sobolev)** achieves `0.6342` **Mean Symmetric DICE** (+1.26% over ANTs, with 90.0% zero-fold regularity). - **Sobolev-Preconditioned TVF (`syntx.tvf`)** achieves a **100% win sweep (90/90 wins)** with `0.6445` **Mean Symmetric DICE** (+2.29% over ANTs) and **`0.0000%` grid folding (**$\det(J) > 0$ **strictly positive everywhere)**. - Tensor acceleration reduces deformable 3D volume execution to **`~12-16s` per pair on modern GPU architectures** ($7.5\times - 24\times$ speedup over classical CPU baselines).

All benchmark evaluation protocols, dataset preparation procedures, and interactive metrology tools are fully documented and reproducible via `docs/run_mb_eval.md`.

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1. Mathematical Framework & Variational Principles

1.1 Differential Geometry of Diffeomorphisms

Let $\Omega \subset \mathbb{R}^d$ ($d \in \{2, 3\}$) denote a compact, connected spatial domain with smooth boundary $\partial\Omega$. Let $I_F, I_M : \Omega \to \mathbb{R}$ represent fixed (target) and moving (source) scalar intensity images. The objective of image registration is to find a spatial transformation $\Phi : \Omega \to \Omega$ such that the deformed moving image $I_M \circ \Phi$ aligns structurally with I_F.

Symbol	Mathematical Definition
$\Omega \subset \mathbb{R}^d$	Continuous physical image domain with coordinates $\mathbf{x} = (x_1, \dots, x_d)^T$
$I_F, I_M \in L^2(\Omega)$	Fixed target and moving source intensity distributions
$\mathrm{Diff}(\Omega)$	Infinite-dimensional Lie group of smooth C^∞ diffeomorphisms on Ω
$\mathfrak{g} = \mathfrak{X}(\Omega)$	Lie algebra of smooth Eulerian velocity vector fields $\mathbf{v} : \Omega \to \mathbb{R}^d$
$\Phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$	Spatial transformation mapping with displacement field $\mathbf{u} : \Omega \to \mathbb{R}^d$
$J_\Phi(\mathbf{x}) = \det(\nabla\Phi(\mathbf{x}))$	Local Jacobian determinant measuring volumetric scaling and orientation
$\Omega_{1/2}$	Symmetrized virtual midpoint domain (Fréchet geodesic mean)
$\phi_{l2r}, \phi_{r2l} \in \mathrm{Diff}(\Omega)$	Forward and reverse half-geodesic transformation trajectories
$\mathcal{D}(I_F, I_M \circ \Phi)$	Image similarity functional (e.g. Local Normalized Cross-Correlation)
$\mathcal{R}(\mathbf{v})$	Regularization functional on the Lie algebra $\mathfrak{g}$ (Sobolev or Gaussian metric)
$\mathcal{G} = (I - \alpha\Delta)^{-s}$	Green's operator associated with the H^s Sobolev Hilbert space inner product

To guarantee topology preservation—precluding tearing, self-intersection, and non-physical singularity formation—the transformation Φ must be an element of the infinite-dimensional Lie group of smooth diffeomorphisms $\mathrm{Diff}(\Omega)$.

1. The Diffeomorphic Condition and the Jacobian Determinant The local orientation and volume preservation of $\Phi \in \text{Diff}(\Omega)$ are governed by its Jacobian matrix $\nabla\Phi(\mathbf{x}) = I + \nabla\mathbf{u}(\mathbf{x})$. The Jacobian determinant is defined as:

$$J_\Phi(\mathbf{x}) = \det(\nabla\Phi(\mathbf{x})) = \det(I + \nabla\mathbf{u}(\mathbf{x}))$$

A transformation Φ is locally invertible, orientation-preserving, and non-singular if and only if:

$$J_\Phi(\mathbf{x}) > 0, \quad \forall \mathbf{x} \in \Omega$$

Regions where $J_\Phi(\mathbf{x}) \leq 0$ represent non-invertible coordinate foldings where the spatial topology collapses.

1.2 Symmetric Normalization (`SyN`) & Geodesic Midpoints

Classical asymmetric registration optimizes $\Phi : \Omega_F \to \Omega_M$, introducing an inherent template bias depending on which volume is designated as target. The Symmetric Normalization (`SyN`) framework [@avants2008symmetric] resolves this by optimizing two diffeomorphic paths $\phi_{l2r}, \phi_{r2l} \in \text{Diff}(\Omega)$ originating from a shared virtual midpoint domain $\Omega_{1/2}$:

$$\phi_{l2r} : \Omega_{1/2} \to \Omega_F, \quad \phi_{r2l} : \Omega_{1/2} \to \Omega_M$$

The full forward mapping is the composite:

$$\Phi_{M\to F} = \phi_{l2r} \circ \phi_{r2l}^{-1}$$

The variational energy functional balances structural image similarity at the midpoint domain with spatial smoothness on the underlying velocity fields:

$$\mathcal{E}(\mathbf{v}_l, \mathbf{v}_r) = \int_{\Omega_{1/2}} \mathcal{S}\left(I_F \circ \phi_{l2r}(\mathbf{x}), I_M \circ \phi_{r2l}(\mathbf{x})\right) d\mathbf{x} + \int_0^1 \left(\|\mathbf{v}_l(t)\|_V^2 + \|\mathbf{v}_r(t)\|_V^2\right) dt$$

where $\|\cdot\|_V$ denotes an admissible Hilbert space norm on $\mathfrak{g}$ enforcing spatial regularity.

1.3 Local Normalized Cross-Correlation (LNCC) Functional

For multi-modal or intra-modality MRI with spatial intensity inhomogeneities (e.g., B_1 field bias), registration optimizes **Local Normalized Cross-Correlation (LNCC)** computed over a localized spatial window $W(\mathbf{x})$ of radius r:

$$\mathcal{L}_{\text{LNCC}}(I_F, I_M) = -\int_\Omega \frac{\left(\int_{W(\mathbf{x})} \tilde{I}_F(\mathbf{y})\tilde{I}_M(\mathbf{y})d\mathbf{y}\right)^2}{\left(\int_{W(\mathbf{x})} \tilde{I}_F^2(\mathbf{y})d\mathbf{y}\right)\left(\int_{W(\mathbf{x})} \tilde{I}_M^2(\mathbf{y})d\mathbf{y}\right)} d\mathbf{x}$$

where $\tilde{I}(\mathbf{y}) = I(\mathbf{y}) - \bar{I}(\mathbf{x})$ denotes local zero-mean intensities over the neighborhood $W(\mathbf{x})$.

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2. Theoretical Analysis & Algorithmic Principles

2.1 Metric Preservation via Single Interpolation

In classical multi-stage registration pipelines (rigid → affine → deformable), intermediate warped intensity images or segmentations are often resampled at each stage. Mathematically, resampling an image I with an interpolation kernel K_σ corresponds to spatial convolution $I_{\text{resampled}} = I * K_\sigma$.

Successive resamplings compound spatial attenuation:

$$I_{\text{final}} = I_0 * K_{\sigma_1} * K_{\sigma_2} * \cdots * K_{\sigma_n}$$

This cumulative low-pass filtering systematically attenuates high-frequency cortical boundaries and gyral sharpness.

The Single Interpolation Principle We enforce exact transformation composition directly in the continuous coordinate manifold:

$$\Phi_{\text{composite}}(\mathbf{x}) = \phi \circ A \circ T_0(\mathbf{x})$$

where $T_0 \in \mathrm{SE}(3)$ is the initial translation, $A \in \mathrm{Aff}(3)$ is the learned affine matrix, and $\phi \in \mathrm{Diff}(\Omega)$ is the non-linear displacement field. The input intensity volume or structural label map is sampled **exactly once** using $\Phi_{\text{composite}}$:

$$I_{\text{warped}}(\mathbf{x}) = I_{\text{native}}\left(\Phi_{\text{composite}}(\mathbf{x})\right)$$

For discrete segmentation maps, the pull-back operation strictly uses nearest-neighbor projection to preserve integer label semantics without artificial partial-volume averaging.

2.2 Functional Gradient Asymptotics & Variance Flooring in LNCC

Let the local cross-correlation coefficient at coordinate $\mathbf{x}$ be written as:

$$r(\mathbf{x}) = \frac{s_{FM}(\mathbf{x})}{\sqrt{s_{FF}(\mathbf{x}) \cdot s_{MM}(\mathbf{x})}}$$

where $s_{FM} = \mathrm{Cov}_W(I_F, I_M)$ and $s_{FF} = \mathrm{Var}_W(I_F), s_{MM} = \mathrm{Var}_W(I_M)$.

Analytical Gradient Singularity The variational L^2 functional derivative with respect to the moving image intensity I_M is:

$$\frac{\delta \mathcal{L}_{\text{LNCC}}}{\delta I_M(\mathbf{x})} = -\frac{2r(\mathbf{x})}{\sqrt{s_{FF}(\mathbf{x}) \cdot s_{MM}(\mathbf{x})}} \left(\tilde{I}_F(\mathbf{x}) - \frac{s_{FM}(\mathbf{x})}{s_{MM}(\mathbf{x})} \tilde{I}_M(\mathbf{x}) \right)$$

In homogeneous anatomical regions (such as cerebral white matter, ventricles, or background zero padding), the local variance vanishes:

$$\lim_{s_{MM} \to 0} \frac{\delta \mathcal{L}_{\text{LNCC}}}{\delta I_M} \sim \mathcal{O}\left(s_{MM}^{-1/2}\right) \to \infty$$

This variance singularity produces unbounded gradient forces in flat regions, causing high-frequency coordinate tearing that directly drives non-diffeomorphic grid folding ($\det(J) \leq 0$).

Safe Variance Floor Regularization To regularize the functional derivative, we impose a lower bound on local variance prior to denominator evaluation:

$$\mathrm{Var}_{\text{safe}}(I) = \max\left(\mathrm{Var}(I), \epsilon\right), \quad \epsilon = 10^{-6}$$

$$\mathcal{L}_{\text{LNCC}}^{\text{safe}} = -\frac{\mathrm{Cov}(I_F, I_M)}{\sqrt{\mathrm{Var}_{\text{safe}}(I_F) \cdot \mathrm{Var}_{\text{safe}}(I_M)}}$$

Coupled with Cauchy-Schwarz bound clamping ($r \in [-1, 1]$), safe variance flooring guarantees bounded gradient trajectories:

$$\left\| \frac{\delta \mathcal{L}_{\text{LNCC}}^{\text{safe}}}{\delta I} \right\|_{\infty} \leq \frac{2}{\epsilon}$$

eliminating derivative spikes across uniform tissue regions.

2.3 Lie Algebra $\mathfrak{so}(3)$ Parameterization & Smooth Exponential Limit

Spatial 3D rotations form the special orthogonal Lie group $\mathrm{SO}(3) = \{R \in \mathbb{R}^{3\times3} : R^T R = I, \det(R) = 1\}$. We parameterize rotations via the Lie algebra $\mathfrak{so}(3)$ of skew-symmetric matrices using the angular velocity vector $\omega = (\omega_x, \omega_y, \omega_z)^T \in \mathbb{R}^3$.

The exponential mapping $\exp : \mathfrak{so}(3) \to \mathrm{SO}(3)$ is evaluated via the closed-form Rodrigues formula:

$$R(\omega) = I + \frac{\sin\theta}{\theta}[\omega]_\times + \frac{1-\cos\theta}{\theta^2}[\omega]_\times^2, \quad \text{where } \theta = \|\omega\|_2$$

and $[\omega]_\times$ is the skew-symmetric cross-product matrix:

$$[\omega]_\times = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}$$

First-Order Taylor Limit at Identity At identity initialization ($\omega = \mathbf{0} \implies \theta = 0$), discrete conditional branching (e.g. `if theta == 0: return I`) creates non-differentiable step boundaries that zero-out automatic-differentiation gradients: $\frac{\partial R}{\partial \omega}\big|_{\omega=\mathbf{0}} = \mathbf{0}$.

We establish a continuous first-order Taylor expansion for $\theta^2 < 10^{-16}$:

$$\lim_{\theta\to0}\frac{\sin\theta}{\theta} = 1, \quad \lim_{\theta\to0}\frac{1-\cos\theta}{\theta^2} = \frac{1}{2}$$

$$R_{\text{approx}}(\omega) = I + [\omega]_\times$$

This analytical limit preserves non-zero linear gradient flow ($\frac{\partial R_{\text{approx}}}{\partial \omega} \neq \mathbf{0}$) directly from epoch 0.

2.4 Physical Anisotropy Scaling & Coordinate Dualization

Let $\mathbf{s} = (s_z, s_y, s_x)$ denote the physical voxel spacing in millimeters. In continuous physical space Ω, coordinate differential operators depend on the physical metric tensor $g_{ij} = \mathrm{diag}(s_z^{-2}, s_y^{-2}, s_x^{-2})$.

When backpropagating functional losses through normalized coordinate grids $\mathbf{x}_{\text{norm}} \in [-1, 1]^d$, chain-rule gradients $\frac{\partial \mathcal{L}}{\partial \mathbf{x}_{\text{norm}}}$ represent dimensionless quantities. Transforming these into physical velocity fields (1/mm) requires dimension-aware metric scaling:

$$\mathbf{s}_{\text{phys}} = \mathrm{flip}\left(\frac{(\mathbf{N}-1)\odot\mathbf{s}}{2}, \dim = 0\right)$$

where $\mathbf{N} = (N_z, N_y, N_x)$ is the grid dimension. The dimension reversal accounts for the indexing convention parity between matrix storage orders (Z, Y, X) and physical Cartesian displacement channels (x, y, z).

2.5 Deterministic Multi-Start Global Search on Transformation Manifolds

Standard gradient descent on the affine Lie group Aff(3) readily converges to local suboptimal basins when subject brains exhibit angular pitch or yaw tilt relative to standard coordinate space.

We formulate a deterministic multi-start search protocol over SE(3) prior to continuous optimization: 1. **Geometric Translation Matching**: Evaluates Center-of-Mass ($\mathbf{t}_{\text{CoM}}$) and Field-of-View geometric centers ($\mathbf{t}_{\text{FOV}}$). 2. **18-Cone Lie Algebra Perturbation Lattice**: Evaluates 18 deterministic rotational candidates:

$$\omega_{\text{candidate}} \in \{\pm4^\circ, \pm8^\circ, \pm12^\circ\} \times \{\mathbf{e}_{\text{pitch}}, \mathbf{e}_{\text{roll}}, \mathbf{e}_{\text{yaw}}\}$$

3. **Foreground Union-Masked Mutual Information**: Scored via deterministic regular spatial sampling over the joint foreground domain $\Omega_{\text{fg}} = \{\mathbf{x} : I_F(\mathbf{x}) > 0.01 \vee I_M(\mathbf{x}) > 0.01\}$:

$$\text{MI}(I_F, I_M; \Omega_{\text{fg}}) = H(I_F) + H(I_M) - H(I_F, I_M)$$

4. **Continuous Multi-Resolution Refinement**: Refines the optimal candidate through multi-scale gradient descent.

2.6 Symmetrical Midpoint Geodesics & Antisymmetric Velocity Projection

In the Symmetric Normalization (SyN) architecture, velocity updates δ_l and δ_r are computed independently from similarity gradients at the virtual midpoint $\Omega_{1/2}$. Unconstrained independent updates allow common-mode translational drift to accumulate, shifting the midpoint domain away from the Fréchet mean of the two images.

Orthogonal Decomposition of Velocity Updates The product Lie algebra $\mathfrak{g} \times \mathfrak{g}$ decomposes into an orthogonal direct sum of antisymmetric (geodesic) and symmetric (drift) subspaces:

$$(\delta_l, \delta_r) = \underbrace{\frac{1}{2}(\delta_l - \delta_r,\ \delta_r - \delta_l)}_{\text{Antisymmetric (Geodesic Velocity)}} + \underbrace{\frac{1}{2}(\delta_l + \delta_r,\ \delta_l + \delta_r)}_{\text{Symmetric (Common-Mode Drift)}}$$

The symmetric component represents pure domain translation through deformation space that does not contribute to image alignment.

We apply the exact orthogonal projection onto the antisymmetric manifold:

$$\mathbf{e}_{\text{drift}} = \delta_l + \delta_r$$

$$\delta_l \leftarrow \delta_l - \frac{1}{2}\mathbf{e}_{\text{drift}}, \quad \delta_r \leftarrow \delta_r - \frac{1}{2}\mathbf{e}_{\text{drift}}$$

This guarantees $\delta_l + \delta_r = \mathbf{0}$ at every iteration, anchoring the virtual domain $\Omega_{1/2}$ strictly at the Fréchet mean.

2.7 Sub-Voxel Metric Symmetry & Anderson-Accelerated Inversion

True diffeomorphic mapping requires the forward transform ϕ_{fwd} and inverse transform ϕ_{inv} to satisfy the involution identity:

$$\phi_{\text{inv}}(\mathbf{x} + \mathbf{u}_{\text{fwd}}(\mathbf{x})) + \mathbf{u}_{\text{fwd}}(\mathbf{x}) = \mathbf{0}, \quad \forall \mathbf{x} \in \Omega$$

Fixed-point Picard iteration $\mathbf{u}_{\text{inv}}^{k+1} = -\mathbf{u}_{\text{fwd}}(\mathbf{x} + \mathbf{u}_{\text{inv}}^k)$ diverges when local strain $\|\nabla \mathbf{u}\| > 1$. We incorporate **Anderson acceleration** (mixing depth $m = 5$) inside the optimization loop:

$$\mathbf{u}_{\text{inv}}^{k+1} = \sum_{j=0}^{m} \alpha_j^* \mathbf{g}(\mathbf{u}_j^k)$$

where the mixing coefficients α_j^* solve the unconstrained least-squares residual minimization $\min_{\sum \alpha_j = 1} \| \sum \alpha_j \mathbf{r}_j \|_2$. This yields sub-voxel identity precision (< 0.02 mm mean identity error) across 3D brain volumes.

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3. Time-Varying Velocity Fields (TVF) & Sobolev-Riemannian Optimization

3.1 Geodesic Flows in Large Deformation Diffeomorphic Metric Mapping

In Large Deformation Diffeomorphic Metric Mapping (LDDMM) [@beg2005computing; @dupuis1998variational; @trouve1998diffeomorphisms; @younes2010shapes], deformations are generated by continuous integration of time-dependent velocity vector fields $\mathbf{v}(t, \mathbf{x}) \in \mathfrak{g}$:

$$\frac{d\phi(t, \mathbf{x})}{dt} = \mathbf{v}(t, \phi(t, \mathbf{x})), \quad \phi(0, \mathbf{x}) = \mathbf{x}, \quad t \in [0, 1]$$

The kinetic energy of the flow is defined by the action integral over the admissible Hilbert space V:

$$E(\mathbf{v}) = \frac{1}{2}\int_0^1 \|\mathbf{v}(t)\|_V^2 \, dt = \frac{1}{2}\int_0^1 \langle \mathcal{L}\mathbf{v}(t), \mathbf{v}(t)\rangle_{L^2} \, dt$$

where $\mathcal{L} = (I - \alpha\Delta)^s$ is a self-adjoint differential operator enforcing spatial smoothness.

Discretized Trajectory Representation The continuous velocity flow is parameterized via T keyframe velocity tensors $\{\mathbf{v}(t_k)\}_{k=0}^{T-1}$ with continuous Catmull-Rom cubic spline interpolation in time. Path consistency is enforced by evaluating the multi-point variational functional across trajectory timepoints:

$$\mathcal{L}_{\mathrm{TVF}} = \frac{1}{3}\left(\mathcal{L}_{\mathrm{LNCC}}(I_F \circ \phi(0), I_M) + \mathcal{L}_{\mathrm{LNCC}}(I_F \circ \phi(0.5), I_M \circ \phi(0.5)^{-1}) + \mathcal{L}_{\mathrm{LNCC}}(I_F, I_M \circ \phi(1))\right)$$

3.2 The Variance Division Singularity in Pointwise Adaptive Optimizers

Standard first- and second-moment adaptive optimizers (such as Adam) compute coordinate-wise parameter updates:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)g_t, \quad v_t = \beta_2 v_{t-1} + (1-\beta_2)g_t^2$$

$$\Delta\mathbf{v}_{\mathrm{raw}} = \frac{m_t/(1-\beta_1^t)}{\sqrt{v_t/(1-\beta_2^t)} + \epsilon}$$

The Metric Collapse on Function Spaces While effective in finite-dimensional machine learning, pointwise adaptive normalization introduces severe pathologies when applied to infinite-dimensional velocity fields $\mathbf{v} \in \mathfrak{g}$: 1. **Noise Amplification in Homogeneous Regions**: In regions where image gradients vanish ($g_t \to 0$), the second moment $v_t \to 0$. Pointwise division by $\sqrt{v_t}$ rescales infinitesimal noise up to $\mathcal{O}(1)$ unit steps. 2. **Destruction of Sobolev Regularity**: Pointwise division destroys spatial correlation between adjacent coordinate grid voxels, injecting high-frequency coordinate shearing into the velocity field. 3. **Momentum Noise Accumulation**: Unregularized moment buffers m_t continuously integrate and reinject high-frequency noise, forcing local Jacobian determinants to collapse ($\det(J) \leq 0$).

3.3 Sobolev-Riemannian Step Preconditioning & CFL Step Bounding (`SobolevAdam`)

To reconcile the adaptive learning rate advantages of moment-based optimization with the strict smoothness requirements of diffeomorphic flows, we formulate optimization on the Sobolev Hilbert space $H^s(\Omega; \mathbb{R}^d)$ equipped with the inner product:

$$\langle \mathbf{u}, \mathbf{w}\rangle_{H^s} = \int_\Omega \langle (I - \alpha\Delta)^s \mathbf{u}(\mathbf{x}), \mathbf{w}(\mathbf{x})\rangle d\mathbf{x}$$

The Riesz representation theorem establishes that the Riemannian gradient $\nabla_{H^s}\mathcal{E}$ with respect to the H^s metric is obtained by applying the Green's operator $\mathcal{G}_{\mathrm{Sobolev}} = (I - \alpha\Delta)^{-s}$ to the standard L^2 functional gradient:

$$\nabla_{H^s}\mathcal{E} = \mathcal{G}_{\mathrm{Sobolev}}\left[\nabla_{L^2}\mathcal{E}\right]$$

1. The `SobolevAdam` Step Operator `SobolevAdam` applies the Sobolev Green's operator **directly to the post-Adam parameter update step $\Delta\mathbf{v}_{\mathrm{raw}}$**:

$$\Delta\mathbf{v}_{\mathrm{smooth}} = \mathcal{G}_{\mathrm{Sobolev}}\left[\Delta\mathbf{v}_{\mathrm{raw}}\right] = \mathcal{F}^{-1}\left(\frac{\mathcal{F}[\Delta\mathbf{v}_{\mathrm{raw}}](\mathbf{k})}{(1+\alpha\|\mathbf{k}\|^2)^s}\right)$$

2. Adaptive Courant-Friedrichs-Lewy (CFL) Step Bounding While Sobolev smoothing ensures spatial differentiability $\mathbf{v} \in C^1(\Omega)$, large discrete displacement updates can still violate the discrete time step condition during forward Euler ODE integration ($\Phi_{k+1} = \Phi_k + \Delta t \cdot \mathbf{v}(\Phi_k)$). To mathematically guarantee that no adjacent spatial coordinates cross over within a discrete time increment, `SobolevAdam` enforces an adaptive **Courant-Friedrichs-Lewy (CFL) step bound**:

$$\mathbf{s}_{\mathrm{CFL}}(\mathbf{x}) = \Delta\mathbf{v}_{\mathrm{smooth}}(\mathbf{x}) \cdot \min\left(1.0, \frac{\mathrm{CFL}_{\max}}{\max_{\mathbf{y}} \|\Delta\mathbf{v}_{\mathrm{smooth}}(\mathbf{y})\|_2 / \Delta x_{\min}}\right)$$

$$\mathbf{v}_{t+1} = \mathbf{v}_t - \eta \cdot \mathbf{s}_{\mathrm{CFL}}$$

where $\mathrm{CFL}_{\max} = 0.35$ voxels and $\Delta x_{\min} = \min(\mathrm{spacing})$.

By coupling H^2 Sobolev metric preconditioning with adaptive CFL displacement step bounding, `SobolevAdam` achieves **strict `0.0000%` grid folding with** $\min\det(J) \geq +0.0517$ **strictly positive everywhere** across all standard and difficult cross-site cohorts.

3.4 Fast 3D Real-FFT Filtering & Memory Pre-Caching

In high-resolution 3D medical image registration ($256\times256\times160$), computing multi-dimensional Fast Fourier Transforms at every optimization iteration can introduce computational latency if spatial grid dimensions are non-factorizable.

We optimize 3D Sobolev smoothing through two computational innovations: 1. **Fourier Green's Operator Pre-Caching (`_SOBOLEV_FILTER_CACHE`)**: The discrete frequency filter $\hat{\mathcal{G}}(\mathbf{k}) = (1+\alpha\|\mathbf{k}\|^2)^{-s}$ depends purely on spatial grid shape and physical voxel spacing. By computing and caching $\hat{\mathcal{G}}(\mathbf{k})$ in device VRAM during the initial epoch of each multi-resolution pyramid level, repeated filter allocations are completely eliminated. 2. **Native Composite Radix-2 Dimensions**: Conventional reflection padding introduces arbitrary boundary dimensions (such as $176\times272\times272$) containing large prime factors (e.g. 11, 17) that degrade FFT performance. Operating directly on native composite dimensions with periodic boundary conditions accelerates 3D Sobolev filtering by $6.5\times$ **per smoothing call**, reducing total 3D registration runtime by $> 40\times$.

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4. Empirical Evaluation on the 90-Pair Mindboggle-101 Benchmark

4.1 Benchmark Protocol, Evaluation Metrics & Canonical Affine Locking

Registration performance is evaluated across **90 3D T1-weighted brain volume pairs** (40 intra-subject longitudinal test-retest pairs and 50 inter-subject cross-site pairs) sampled from the standardized Mindboggle-101 cohort [@klein2012101; @klein2017mindboggle].

1. Evaluation Metrics Registration quality is benchmarked using utility-computed quantitative metrics: 1. **Bidirectional Cortical DKT31 Overlap**: Evaluated symmetrically across 62 discrete manual cortical labels (31 labels per hemisphere) using nearest-neighbor pull-back (`interpolator='nearestNeighbor'`):

$$\mathrm{Dice}_{\mathrm{fix}} = \frac{2|\mathbf{L}_{\mathrm{fix}} \cap (\mathbf{L}_{\mathrm{mov}} \circ \phi_{\mathrm{fwd}})|}{|\mathbf{L}_{\mathrm{fix}}| + |\mathbf{L}_{\mathrm{mov}} \circ \phi_{\mathrm{fwd}}|}, \quad \mathrm{Dice}_{\mathrm{mov}} = \frac{2|\mathbf{L}_{\mathrm{mov}} \cap (\mathbf{L}_{\mathrm{fix}} \circ \phi_{\mathrm{inv}})|}{|\mathbf{L}_{\mathrm{mov}}| + |\mathbf{L}_{\mathrm{fix}} \circ \phi_{\mathrm{inv}}|}$$

$$\mathrm{Dice}_{\mathrm{sym}} = \frac{1}{2}\left(\mathrm{Dice}_{\mathrm{fix}} + \mathrm{Dice}_{\mathrm{mov}}\right)$$

2. **Diffeomorphic Manifold Regularity**: Evaluates non-invertible spatial grid folding percentage and minimum Jacobian determinant:

$$\text{Fold\%} = \frac{1}{|\Omega|} \int_{\Omega} \mathbf{1}_{(\det(J(\mathbf{x})) \leq 0)} \, d\mathbf{x} \times 100\%, \quad \min_{\mathbf{x} \in \Omega} \det(J(\mathbf{x}))$$

3. **Physical Inverse Identity Consistency (mm)**: Real physical coordinate residual map:

$$\mathbf{e}(\mathbf{x}) = \|\phi_{\text{inv}}(\mathbf{x} + \mathbf{u}_{\text{fwd}}(\mathbf{x})) + \mathbf{u}_{\text{fwd}}(\mathbf{x})\|_2 \quad \text{(in mm)}$$

Reporting Mean Inverse Error, 95th Percentile (p_{95}), and Peak Maximum Error. 4. **Execution Runtime**: Total wall-clock fit time in seconds.

2. Canonical Affine Locking To guarantee that performance differences isolate non-linear deformation mechanics, **all algorithms share the exact same pre-computed canonical affine transform** (`results/canonical_affines/pair_XXX_affine.mat`, `0.3499` baseline DICE). None of the methods alter or continue affine optimization during deformable registration.

4.2 Parameter Parity Across Evaluated Algorithms

All registration arms adhere to standardized parameter conventions matching the canonical evaluation protocol:

Parameter	ANTs C++ SyN (CPU Baseline)	Eulerian SyN (Gaussian)	Eulerian SyN (Sobolev)	CFL-`SobolevAdam` TVF (Peak)
Formulation	Eulerian SyN	Eulerian SyN	Eulerian SyN	Continuous TVF (ODE)
Similarity Metric	LNCC ($5 \times 5 \times 5$, `cc2`)	LNCC ($5 \times 5 \times 5$, `cc2`)	LNCC ($5 \times 5 \times 5$, `cc2`)	3-Point LNCC ($t \in [0, 0.5, 1]$)
Gradient Step / LR	`0.25`	`0.25`	`0.25`	`1.20` (`SobolevAdam`)
CFL Step Bound	N/A	N/A	N/A	`max_step_norm = 0.35` voxels
Fluid Smoothing (σ_f)	$\sigma^2 = 3.0$ ($\sigma = 1.732$ mm)	$\sigma^2 = 3.0$ (ITK Bessel)	Fourier Sobolev ($H^{1.5}$)	Fluid Velocity $\sigma_f = 1.0$
Elastic Smoothing (σ_e)	`0.0` (pure fluid)	`0.0` (pure fluid)	`0.0` (pure fluid)	$\sigma_e = 0.035$, $\alpha_{\text{sob}} = 0.035$ mm^{-1}
Multi-Scale Pyramid	`[100, 100, 50]`	`[100, 100, 50]`	`[100, 100, 50]`	`[100, 50, 10]` (Peak) / `[100, 40, 0]` (Fast)
Inverse Solver	In-loop fixed point	In-loop Anderson ($m = 5$)	In-loop Anderson ($m = 5$)	Reverse ODE flow
Initial Transform	Locked Canonical Affine	Locked Canonical Affine	Locked Canonical Affine	Locked Canonical Affine

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4.3 Aggregate 90-Pair Performance Results

Algorithm	Win Rate vs ANTs	Mean Sym DICE	Fixed DICE	Moving DICE	Fold (%)
ANTs C++ SyN (CPU Baseline)	Baseline (0/90)	`0.6216`	`0.6218`	`0.6214`	`0.0000%`
Eulerian SyN (Gaussian)	88 / 90 (97.8%)	**`0.6382` (+1.66%)**	`0.6385`	`0.6379`	`0.0010%`
Eulerian SyN (Sobolev)	81 / 90 (90.0%)	**`0.6342` (+1.26%)**	`0.6345`	`0.6339`	`0.0000%`
`SobolevAdam` TVF (Peak)	**90 / 90 (100.0%)**	**`0.6445` (+2.29%)**	**`0.6449`**	**`0.6441`**	**`0.0000%`**

4.4 Multi-Scale Schedule Progression & Demographic Asymmetry Analysis

To quantify the resolution-scale trade-offs between optimization latency and deformation accuracy, we evaluate three standardized multi-resolution schedules across both intra-cohort (Pair 00: OASIS-16 → OASIS-17) and severe cross-site demographic mismatch (**`mbhard`** / Pair 77: OASIS-8 elderly → NKI-3 young adult) using CFL-`SobolevAdam`:

Dataset Pair	Multi-Scale Schedule	Sym DICE	Fixed Space DICE	Moving Space DICE	Grid Folds (%)	Min $\det(J)$	Execution Time	Diffeomorphic Status
Pair 00 (OASIS-16 → OASIS-17)	`[100, 40, 0]`	`0.5857`	`0.5624`	`0.6090`	`0.0000%`	`+0.1445`	`51.8s`	Fold-Free (PASS)
Pair 00 (OASIS-16 → OASIS-17)	`[100, 50, 10]`	**`0.6370`**	`0.6174`	`0.6566`	**`0.0000%`**	**`+0.0753`**	**`169.9s`**	**Peak Production (PASS)**
Pair 00 (OASIS-16 → OASIS-17)	`[100, 50, 20]`	**`0.6466`**	`0.6289`	`0.6643`	`0.0001%`	`0.0000`	`308.4s`	Trace Folds
`mbhard` (OASIS-8 → NKI-3)	`[100, 40, 0]`	**`0.5784`**	`0.6059`	`0.5508`	**`0.0000%`**	**`+0.1231`**	**`39.9s`**	Fold-Free (PASS)

Dataset Pair	Multi-Scale Schedule	Sym DICE	Fixed Space DICE	Moving Space DICE	Grid Folds (%)	Min $\det(J)$	Execution Time	Diffeomorphic Status
`mbhard` (OASIS-8 → NKI-3)	[100, 50, 10]	**0.6126**	**0.6442**	0.5809	**0.0000%**	**+0.0517**	**150.4s**	**Peak Production (PASS)**
`mbhard` (OASIS-8 → NKI-3)	[100, 50, 20]	**0.6158**	**0.6497**	0.5818	0.0002%	0.0000	306.8s	Trace Folds

Demographic Volume Asymmetry in Directional Overlap On `mbhard`, Fixed Space DICE consistently reaches **`0.6442 - 0.6497`**, while Moving Space DICE reaches **`0.5809 - 0.5818`**. This asymmetry reflects fundamental brain morphology: - When warping the atrophic, thin-sulcal elderly moving brain (OASIS-8) into the young adult fixed space (NKI-3), labels expand into thick cortical ribbons, maximizing continuous target overlap. - When warping young adult labels into the narrow sulci of the elderly brain, nearest-neighbor discretization over high-curvature sulcal boundaries introduces a volume-penalization effect. - Standardizing evaluation via **Symmetric Mean DICE** ($\text{Dice}_{\text{sym}} = \frac{1}{2}(\text{Dice}_{\text{fix}} + \text{Dice}_{\text{mov}})$) guarantees unbiased comparison across demographic cohorts.

4.5 Longitudinal vs Cross-Site Performance Breakdown

Cohort Subset	N Pairs	ANTs C++ SyN	Eulerian SyN (Gaussian)	`SobolevAdam` TVF	TVF Advantage
Intra-Subject (Longitudinal)	40	0.6842	0.6985	**0.7048**	**+2.06%**
Inter-Subject (Cross-Site)	50	0.5715	0.5900	**0.5962**	**+2.47%**
Full Cohort Combined	90	0.6216	0.6382	**0.6445**	**+2.29%**

4.6 16 Inter-Study Affine Registration Evaluation

Affine Alignment Protocol	Mean Sym DICE	Speed per Pair	Convergence Rate
Deterministic Multi-Start (`auto`)	**0.3460**	**5.60s**	**100% (16/16)**
ANTsPy `ants_fast`	0.3456	5.48s	100% (16/16)

Affine Alignment Protocol	Mean Sym DICE	Speed per Pair	Convergence Rate
Translation Only (`com_only`)	`0.2434`	**`0.24s`**	100% (16/16)
Single-Start Gradient Descent	`0.2259`	`7.94s`	68.7% (Entrapment prone)

4.7 Computational Latency & Hardware Scalability

Benchmark Metric	ANTs C++ SyN (CPU)	GPU Accelerated (MPS)	High-Throughput GPU (CUDA)
Multi-Start Affine Registration	~28.5 s	~2.8 s	~**1.2 s** (24× speedup)
Deformable SyN (per 3D Pair)	~85–120 s	~24–28 s	~**12–16 s** (7.5× speedup)
90-Pair Cohort Total Time	~2.5–3.0 hours	~40 minutes	~**20–25 minutes**

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5. Diagnostic Metrology & Quality Assurance (`syntx.viz`)

To ensure transparent visual verification, every registration generates a standard 5-figure diagnostic report: 1. **Input Anatomical Verification**: Tri-planar views in canonical LPI space with physical voxel spacing anisotropy scaling. 2. **4-Panel Diagnostic Metrology**: - **Deformed Coordinate Grid**: Continuous coordinate transformation mesh. - **Log-Jacobian Determinant Map**: Divergent seismic colormap centered at 1.0. **Singular folding voxels ($\det(J) \leq 0$) are explicitly highlighted with a solid neon lime green overlay**. - **Physical Inverse Identity Error Map**: Real physical coordinate residual $\|\phi_{\text{inv}}(\mathbf{x} + \mathbf{u}(\mathbf{x})) + \mathbf{u}(\mathbf{x})\|_2$ in millimeters. - **High-Contrast Boundary Overlap**: Canny structural edge contour overlay. 3. **Time-Varying Velocity Quivers**: Continuous velocity magnitude heatmaps with amplified quiver flow vectors and domain Thin-Plate Bending Energy:

$$\text{Bnd}(\mathbf{v}) = \frac{1}{|\Omega|} \int_{\Omega} \left(\|\nabla^2 v_x\|_F^2 + \|\nabla^2 v_y\|_F^2 + \|\nabla^2 v_z\|_F^2 \right) d\mathbf{x}$$

4. **Convergence Diagnostics**: Epoch-by-epoch LNCC loss progression and bidirectional Cortical DICE overlap trajectories.

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6. Reproducibility & Open Science

To foster full experimental transparency, all algorithms, benchmark datasets, evaluation scripts, and interactive metrology tools presented in this work are open-source and structured for single-command replication.

A complete step-by-step tutorial is available in the companion evaluation guide: > **Reproducible Benchmark Guide**:
> `docs/run_mb_eval.md` — **Mindboggle-101 Deformable Registration Benchmark Tutorial**

This tutorial provides end-to-end instructions for: 1. **Environment Configuration**: Automated setup instructions across NVIDIA CUDA GPUs, Apple Silicon MPS, and multi-threaded CPU environments. 2. **Standardized Dataset Organization**: Automated scripts to retrieve and structure the 101 labeled T1-weighted volumes and manual DKT31 cortical label maps from the Mindboggle project. 3. **Single-Pair

Metrology Reproduction: One-command reproduction of the standard 5-figure diagnostic report on challenging cross-site pairs (`mbhard` / Pair 77: OASIS-8 $\rightarrow$ NKI-3). 4. **Full 90-Pair Population Evaluation**: Batch execution scripts that compute bidirectional cortical DKT31 DICE scores, numerical Jacobian singularity rates, real physical inverse consistency errors (in mm), and compile aggregate Markdown/HTML reports.

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7. Conclusion

This work establishes a mathematically grounded, computationally efficient framework for symmetric diffeomorphic image registration. By addressing variance singularities in local similarity functionals, establishing smooth Lie group limits, formulating the `SobolevAdam` Riemannian step preconditioning algorithm, and enforcing antisymmetric geodesic projections, `syntx` eliminates numerical instabilities while achieving a **100% win sweep across all 90 Mindboggle benchmark pairs** (`0.6445` vs `0.6216` Mean Symmetric DICE) with strict diffeomorphic manifold regularity ($\det(J) > 0$). Tensor-accelerated execution provides $7.5\times - 24\times$ speedups over CPU baselines, establishing an open-source, robust foundation for large-scale computational neuroanatomy.

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